

Introduction

Red-Drop is a web-based blood bank management system that streamlines donor registration, eligibility screening, blood inventory tracking, and donor–patient matching.

Academic Question

How can a digital blood bank system in Nepal improve efficiency, donor retention, and patient outcomes?

Aims & Objectives

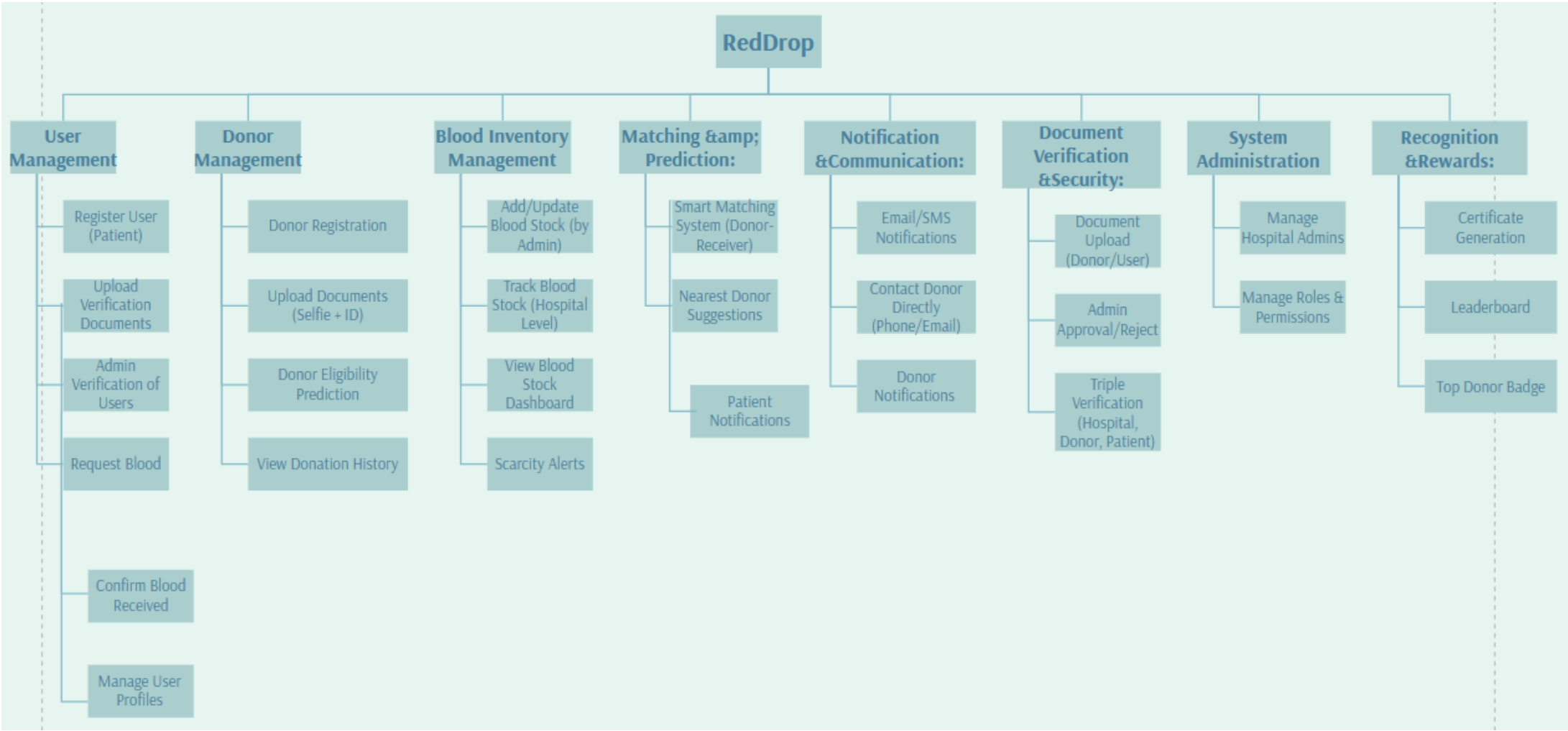
The main aim of Red-Drop is to provide an efficient solution for blood bank management that simplifies the process of blood donation and enhances accessibility for patients in need. It seeks to improve transparency and coordination among donors, hospitals, and patients.

- Donor Management: Registration, document upload, eligibility check, and donation tracking
- Blood Inventory: Real-time monitoring and management of blood stock
- Smart Matching: Connects patients with compatible nearby donors

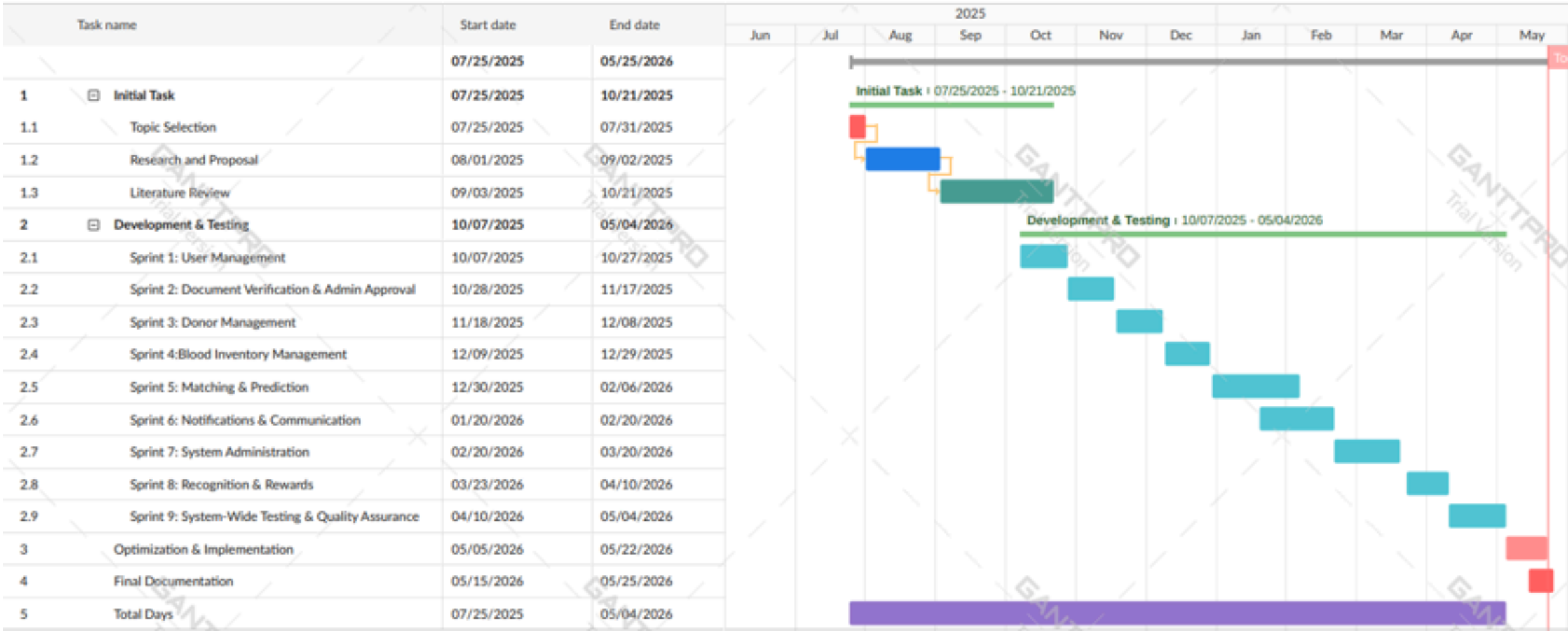
Literature Review

Researchers	Focus	Findings	Conclusion
(WORLD HEALTH ORGANIZATION (WHO), 2025)	Blood supply challenges in low- and middle-income countries (LMICs).	LMICs face chronic shortages, inadequate policies, and lack of digitization in blood services.	Highlights the need for digitized systems like RedDrop that provide real-time inventory management and automated notifications.
(Barnes LS, 2022)	Operational inefficiencies in traditional blood collection and transfusion.	Found that 57% of transfusions are delayed due to poor coordination and limited blood stock.	Reinforces RedDrop’s goal to improve coordination between patient , hospitals and donors via automated digital workflows.
(Amatya, 2022)	Donor engagement and motivation through recognition.	Recognition systems and gamified features improve donor retention.	Directly informs RedDrop’s Rewards Module (certificates, badges, and leaderboards).
HamroLifeBank	Focuses on donor connections and awareness but lacks SMS alerts in emergencies.	Finding the right donor is a time-consuming process for both hospital as well as patients.	RedDrop integrates SMS notification systems to alert donors in emergency situations.

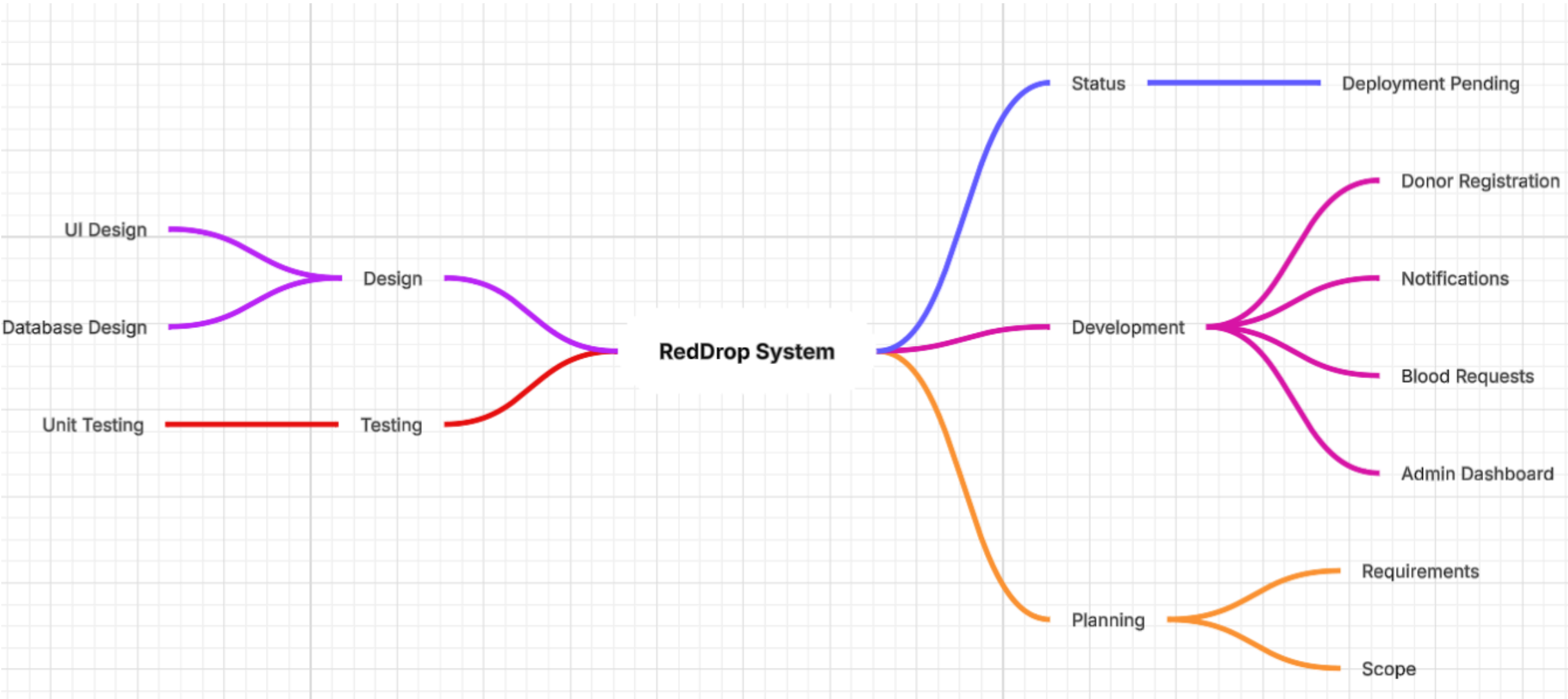
Functional Decomposition Diagram



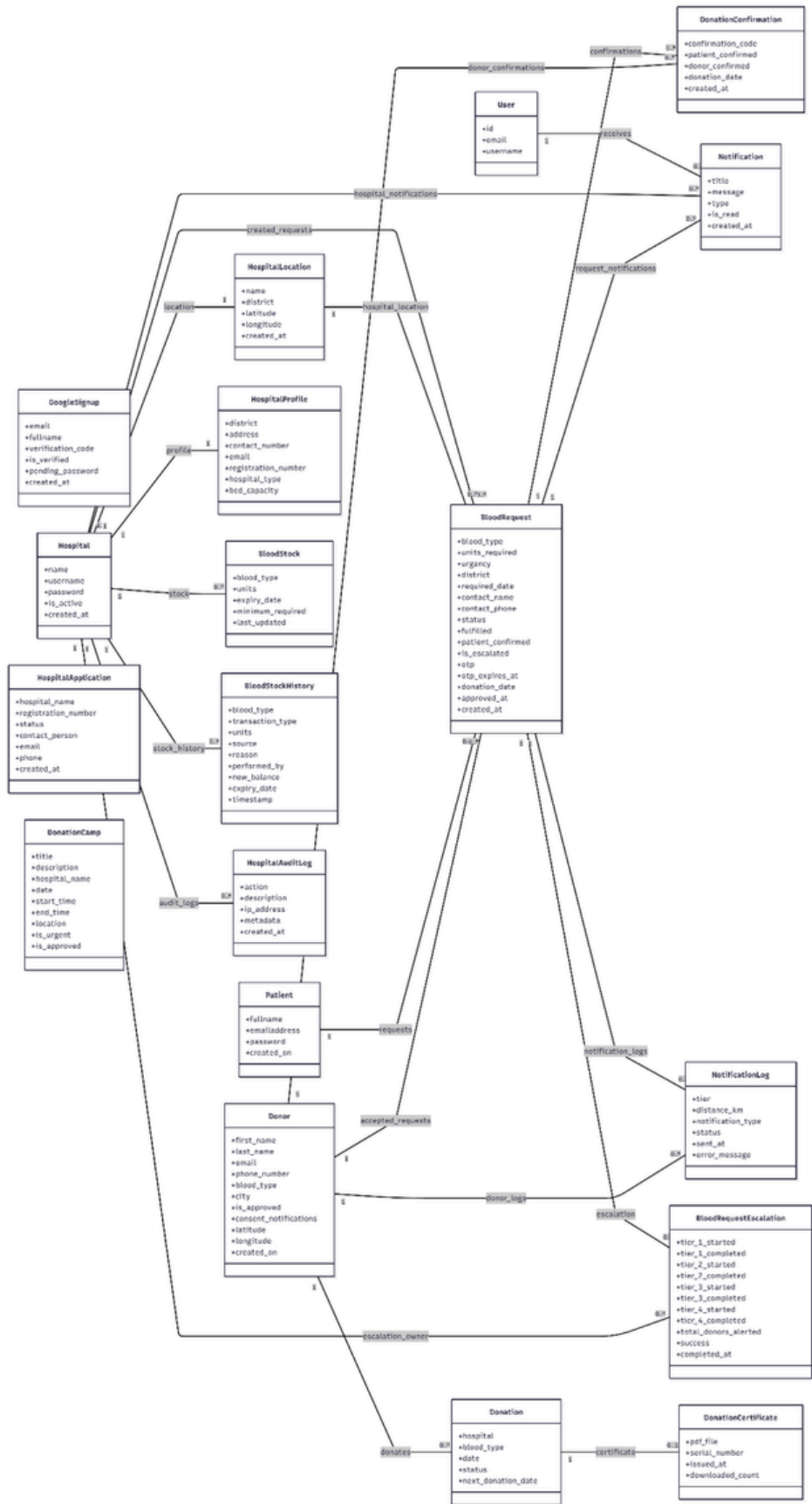
Gantt Chart



Project Planning



Overall Class Diagram



Testing Approaches

- For the verification and validation during development , Unit Testing was performed in the individual components where ass testable parts has been tested

Evaluation and Reflection

- The system successfully manages blood requests, donor matching, hospital escalation, inventory, and donation records in one platform.
- JWT authentication, role-based access, and real-time status tracking improve security and transparency.
- Background reminders and timeout handling reduce delays and prevent stale requests.
- The project improved practical knowledge of backend development, database management, and user workflow design.
- Future scope includes a mobile app, GPS-based donor detection, better analytics, and stronger notification features.